



## Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

### COS20028 Big Data Architecture and Application Lab Task 5 - Semester 2, 2024

**Assessment Title:** Analysing Ad campaign data with Pig.

**Assessment Outcome:** Pass/Fail (for practice)

**Assessable Item:**

- One (1) piece of a report containing the complete Pig Latin codes for preparing the data and finding the analytic result with the screenshot of submitted commands and execution results.

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## Purpose of Assignment

This assignment evaluates whether a student understands how to achieve an analytic goal via Apache Pig Latin.

This is an individual assignment and is mandatory for passing this unit.

## Introduction

The source data is built in the virtual machine image under the path of `/home/training/training_materials/analyst/data`. Our goal is using Apache Pig Latin to find out the **top five (5)** websites that associated with the high cost in **France** from `"ad_data1.txt"` and in **any country** from `"ad_data2.txt"`. The cost for putting up an advertisement is recorded in the `cpc` attribute. Whether an advertisement is clicked is recorded by `"1"` in the `was_clicked` attribute.

The samples and the corresponding attribute name are listed as follows:

**ad\_data1.txt:**

| Keyword     | Champaign_id | Date       | Time      | Display_Site          |
|-------------|--------------|------------|-----------|-----------------------|
| lightweight | D8           | 05/01/2013 | 00:00:08  | gamersite.example.com |
| Was_Clicked | CPC          | Country    | Placement |                       |
| 0           | 72           | USA        | SIDE      |                       |

- The above example tuple shows that this advertisement is in the D8 champaign with the keyword "lightweight", displayed in USA on the website "gamersite.example.com" at 00:00:08 on 05/01/2013 on the side of the webpage. The user didn't click on this advertisement. Otherwise, the company will be charged 72 cents.

**ad\_data2.txt:**

| Campaign_id | Date        | Time     | Display_Site        |
|-------------|-------------|----------|---------------------|
| A1          | 05-01-2013  | 00:00:47 | techwiz.example.com |
| Placement   | Was_Clicked | CPC      | Keyword             |
| SIDE        | 0           | 62       | camera              |

- The above example tuple shows that this advertisement is in the A1 campaign with the keyword “camera”, displayed on the website “techwiz.example.com” at 00:00:47 on 05/01/2013 on the side of the webpage. The user didn’t click on this advertisement. Otherwise, the company will be charged 62 cents.

**Assignment Goal**

This assignment aims at training students to get familiar with Apache Pig Latin for managing data into the specific format for further uses.

**Assignment Task**

The following tasks should be accomplished:

1. Write a set of Pig Latin codes for organising the data from multiple sources into the identical format.
2. Write a Pig Latin code and use file patterns (globs) to load files for analysing the content to find the sum of cost based on the value in the *cpc* attribute. Note that only the advertisement, which is clicked will be charged and have the corresponding cost. If the advertisement is not clicked, it should not be included in the calculation.

Please use the template to prepare your submission.

**Submission Requirement**

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

**Rubric: Lab Task 5**

| Task                                                                                                                                                      | Assessment Outcome |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional. | Pass/Fail          |

----- End of Assignment -----